

Chapter 1

Complex Numbers

Only One Option Correct Type Questions

1. A man walks a distance of 3 units from the origin towards the north-east ($N 45^\circ E$) direction. From there, he walks a distance of 4 units towards the north-west ($N 45^\circ W$) direction to reach a point P . Then the position of P in the Argand plane is [IIT-JEE-2007 (Paper-1)]
- (A) $3e^{i\pi/4} + 4i$ (B) $(3 - 4i)e^{i\pi/4}$
(C) $(4 + 3i)e^{i\pi/4}$ (D) $(3 + 4i)e^{i\pi/4}$
2. If $|z| = 1$ and $z \neq \pm 1$, then all the values of $\frac{z}{1-z^2}$ lie on [IIT-JEE-2007 (Paper-2)]
- (A) A line not passing through the origin (B) $|z| = \sqrt{2}$
(C) The x-axis (D) The y-axis
3. A particle P starts from the point $z_0 = 1 + 2i$, where $i = \sqrt{-1}$. It moves first horizontally away from origin by 5 units and then vertically away from origin by 3 units to reach a point z_1 . From z_1 the particle moves $\sqrt{2}$ units in the direction of the vector $\hat{i} + \hat{j}$ and then it moves through an angle $\frac{\pi}{2}$ in anticlockwise direction on a circle with centre at origin, to reach a point z_2 . The point z_2 is given by [IIT-JEE-2008 (Paper-2)]
- (A) $6 + 7i$ (B) $-7 + 6i$
(C) $7 + 6i$ (D) $-6 + 7i$
4. Let $z = x + iy$ be a complex number where x and y are integers. Then the area of the rectangle whose vertices are the roots of the equation $z\bar{z}^3 + \bar{z}z^3 = 350$ is [IIT-JEE-2009 (Paper-1)]
- (A) 48 (B) 32
(C) 40 (D) 80
5. Let $z = \cos\theta + i \sin\theta$. Then the value of $\sum_{m=1}^{15} \operatorname{Im}(z^{2m-1})$ at $\theta = 2^\circ$ is [IIT-JEE-2009 (Paper-1)]
- (A) $\frac{1}{\sin 2^\circ}$ (B) $\frac{1}{3 \sin 2^\circ}$
(C) $\frac{1}{2 \sin 2^\circ}$ (D) $\frac{1}{4 \sin 2^\circ}$

6. Let z be a complex number such that the imaginary part of z is non-zero and $a = z^2 + z + 1$ is real. Then a **cannot** take the value [IIT-JEE-2012 (Paper-1)]
- (A) -1 (B) $\frac{1}{3}$
(C) $\frac{1}{2}$ (D) $\frac{3}{4}$
7. Let complex numbers α and $\frac{1}{\alpha}$ lie on circles $(x - x_0)^2 + (y - y_0)^2 = r_2$ and $(x - x_0)^2 + (y - y_0)^2 = 4r^2$, respectively. If $z_0 = x_0 + iy_0$ satisfies the equation $2|z_0|^2 = r^2 + 2$, then $|\alpha| =$ [JEE (Adv)-2013 (Paper-1)]
- (A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$
(C) $\frac{1}{\sqrt{7}}$ (D) $\frac{1}{3}$
8. Let S be the set of all complex numbers z satisfying $|z - 2 + i| \geq \sqrt{5}$. If the complex number z_0 is such that $\frac{1}{|z_0 - 1|}$ is the maximum of the set $\left\{ \frac{1}{|z - 1|} : z \in S \right\}$, then the principal argument of $\frac{4 - z_0 - \bar{z}_0}{z_0 - \bar{z}_0 + 2i}$ is [JEE (Adv)-2019 (Paper-1)]
- (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$
(C) $-\frac{\pi}{2}$ (D) $\frac{3\pi}{4}$
9. Let $\theta_1, \theta_2, \dots, \theta_{10}$ be positive valued angles (in radian) such that $\theta_1 + \theta_2 + \dots + \theta_{10} = 2\pi$. Define the complex numbers $z_1 = e^{i\theta_1}, z_k = z_{k-1}e^{i\theta_k}$ for $k = 2, 3, \dots, 10$, where $i = \sqrt{-1}$. Consider the statement P and Q given below: [JEE (Adv)-2021 (Paper-1)]
- $P : |z_2 - z_1| + |z_3 - z_2| + \dots + |z_{10} - z_9| + |z_1 - z_{10}| \leq 2\pi$
 $Q : |z_2^2 - z_1^2| + |z_3^2 - z_2^2| + \dots + |z_{10}^2 - z_9^2| + |z_1^2 - z_{10}^2| \leq 4\pi$
- Then,
- (A) P is **TRUE** and Q is **FALSE** (B) Q is **TRUE** and P is **FALSE**
(C) Both P and Q are **TRUE** (D) Both P and Q are **FALSE**

One or More Option(s) Correct Type Questions

10. Let z_1 and z_2 be two distinct complex numbers and let $z = (1 - t)z_1 + tz_2$ for some real number t with $0 < t < 1$. If $\text{Arg}(w)$ denotes the principal argument of a nonzero complex number w , then [IIT-JEE-2010 (Paper-1)]
- (A) $|z - z_1| + |z - z_2| = |z_1 - z_2|$ (B) $\text{Arg}(z - z_1) = \text{Arg}(z - z_2)$
(C) $\left| \frac{z - z_1}{z_2 - z_1} \cdot \frac{\bar{z} - \bar{z}_1}{\bar{z}_2 - \bar{z}_1} \right| = 0$ (D) $\text{Arg}(z - z_1) = \text{Arg}(z_2 - z_1)$
11. Let ω be a complex cube root of unity with $\omega \neq 1$ and $P = [p_{ij}]$ be a $n \times n$ matrix with $p_{ij} = \omega^{ij}$. Then $P^2 \neq 0$, when $n =$ [JEE (Adv)-2013 (Paper-2)]
- (A) 57 (B) 55
(C) 58 (D) 56

12. Let $\omega = \frac{\sqrt{3}+i}{2}$ and $P = \{\omega^n : n = 1, 2, 3, \dots\}$. Further $H_1 = \left\{z \in \mathbb{C} : \operatorname{Re} z > \frac{1}{2}\right\}$ and $H_2 = \left\{z \in \mathbb{C} : \operatorname{Re} z < \frac{-1}{2}\right\}$, where $\mathbb{C}$ is the set of all complex numbers. If $z_1 \in P \cap H_1$, $z_2 \in P \cap H_2$ and O represents the origin, then $\angle z_1 O z_2 =$

[JEE (Adv)-2013 (Paper-2)]

- (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{6}$
(C) $\frac{2\pi}{3}$ (D) $\frac{5\pi}{6}$

13. Let $a, b \in \mathbb{R}$ and $a^2 + b^2 \neq 0$. Suppose $S = \left\{z \in \mathbb{C} : z = \frac{1}{a+ibt}, t \in \mathbb{R}, t \neq 0\right\}$, where $i = \sqrt{-1}$. If $z = x + iy$ and $z \in S$, then (x, y) lies on

[JEE (Adv)-2016 (Paper-2)]

- (A) The circle with radius $\frac{1}{2a}$ and centre $\left(\frac{1}{2a}, 0\right)$ for $a > 0, b \neq 0$
(B) The circle with radius $-\frac{1}{2a}$ and centre $\left(-\frac{1}{2a}, 0\right)$ for $a < 0, b \neq 0$
(C) The x-axis for $a \neq 0, b = 0$
(D) The y-axis for $a = 0, b \neq 0$

14. Let a, b, x and y be real numbers such that $a - b = 1$ and $y \neq 0$. If the complex number $z = x + iy$ satisfies $\operatorname{Im}\left(\frac{az+b}{z+1}\right) = y$, then which of the following is(are) possible value(s) of x ? [JEE (Adv)-2017 (Paper-1)]

- (A) $-1 + \sqrt{1-y^2}$ (B) $1 - \sqrt{1+y^2}$
(C) $1 + \sqrt{1+y^2}$ (D) $-1 - \sqrt{1-y^2}$

15. For a non-zero complex number z , let $\arg(z)$ denote the principal argument with $-\pi < \arg(z) \leq \pi$. Then, which of the following statement(s) is (are) FALSE? [JEE (Adv)-2018 (Paper-1)]

- (A) $\arg(-1-i) = \frac{\pi}{4}$, where $i = \sqrt{-1}$
(B) The function $f : \mathbb{R} \rightarrow (-\pi, \pi]$, defined by $f(t) = \arg(-1+it)$ for all $t \in \mathbb{R}$, is continuous at all points of $\mathbb{R}$, where $i = \sqrt{-1}$
(C) For any two non-zero complex number z_1 and z_2 , $\arg\left(\frac{z_1}{z_2}\right) - \arg(z_1) + \arg(z_2)$ is an integer multiple of 2π
(D) For any three given distinct complex numbers z_1, z_2 and z_3 , the locus of the point z satisfying the condition $\arg\left(\frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)}\right) = \pi$, lies on a straight line

16. Let s, t, r be non-zero complex numbers and L be the set of solutions $z = x + iy$ ($x, y \in \mathbb{R}, i = \sqrt{-1}$) of the equation $sz + t\bar{z} + r = 0$, where $\bar{z} = x - iy$. Then, which of the following statement(s) is (are) TRUE?

[JEE (Adv)-2018 (Paper-2)]

- (A) If L has exactly one element, then $|s| \neq |t|$
(B) If $|s| = |t|$, then L has infinitely many elements
(C) The number of elements in $L \cap \{z : |z - 1 + i| = 5\}$ is at most 2
(D) If L has more than one element, then L has infinitely many elements

17. Let S be the set of all complex numbers z satisfying $|z^2 + z + 1| = 1$. Then which of the following statements is/are TRUE?
[JEE (Adv)-2020 (Paper-1)]

(A) $\left|z + \frac{1}{2}\right| \leq \frac{1}{2}$ for all $z \in S$

(B) $|z| \leq 2$ for all $z \in S$

(C) $\left|z + \frac{1}{2}\right| \geq \frac{1}{2}$ for all $z \in S$

(D) The set S has exactly four elements

18. For any complex number $w = c + id$, let $\arg(w) \in (-\pi, \pi]$, where $i = \sqrt{-1}$. Let α and β be real numbers such that for all complex numbers $z = x + iy$ satisfying $\arg\left(\frac{z + \alpha}{z + \beta}\right) = \frac{\pi}{4}$, the ordered pair (x, y) lies on the circle

$$x^2 + y^2 + 5x - 3y + 4 = 0$$

[JEE (Adv)-2021 (Paper-1)]

Then which of the following statements is (are) TRUE?

(A) $\alpha = -1$

(B) $\alpha\beta = 4$

(C) $\alpha\beta = -4$

(D) $\beta = 4$

19. Let $\bar{z}$ denote the complex conjugate of a complex number z . If z is a non-zero complex number for which both real and imaginary parts of $(\bar{z})^2 + \frac{1}{z^2}$ are integers, then which of the following is/are possible value(s) of $|z|$?
[JEE (Adv)-2022 (Paper-2)]

(A) $\left(\frac{43 + 3\sqrt{205}}{2}\right)^{1/4}$

(B) $\left(\frac{7 + \sqrt{33}}{4}\right)^{1/4}$

(C) $\left(\frac{9 + \sqrt{65}}{4}\right)^{1/4}$

(D) $\left(\frac{7 + \sqrt{13}}{6}\right)^{1/4}$

Linked Comprehension Type Questions

Paragraph for Q.Nos. 19 to 21

Let A, B, C be three sets of complex numbers as defined below

$$A = \{z : \operatorname{Im} z \geq 1\}$$

$$B = \{z : |z - 2 - i| = 3\}$$

$$C = \{z : \operatorname{Re}((1-i)z) = \sqrt{2}\}.$$

[IIT-JEE-2008 (Paper-1)]

Choose the correct answer :

20. The number of elements in the set $A \cap B \cap C$ is

(A) 0

(B) 1

(C) 2

(D) ∞

21. Let z be any point in $A \cap B \cap C$. The $|z + 1 - i|^2 + |z - 5 - i|^2$ lies between

(A) 25 and 29

(B) 30 and 34

(C) 35 and 39

(D) 40 and 44

22. Let z be any point in $A \cap B \cap C$ and let w be any point satisfying $|w - 2 - i| < 3$. Then, $|z| - |w| + 3$ lies between
- (A) -6 and 3 (B) -3 and 6
 (C) -6 and 6 (D) -3 and 9

Paragraph for Q.Nos. 22 and 23

Let $S = S_1 \cap S_2 \cap S_3$, where

$$S_1 = \{z \in \mathbb{C} : |z| < 4\},$$

$$S_2 = \left\{ z \in \mathbb{C} : \operatorname{Im} \left[\frac{z-1+\sqrt{3}i}{1-\sqrt{3}i} \right] > 0 \right\} \text{ and}$$

$$S_3 = \{z \in \mathbb{C} : \operatorname{Re} z > 0\}.$$

[JEE (Adv)-2013 (Paper-2)]

Choose the correct answer :

23. Area of $S =$

- (A) $\frac{10\pi}{3}$ (B) $\frac{20\pi}{3}$
 (C) $\frac{16\pi}{3}$ (D) $\frac{32\pi}{3}$

24. $\min_{z \in S} |1 - 3i - z| =$

- (A) $\frac{2-\sqrt{3}}{2}$ (B) $\frac{2+\sqrt{3}}{2}$
 (C) $\frac{3-\sqrt{3}}{2}$ (D) $\frac{3+\sqrt{3}}{2}$

Matrix-Match / Matrix Based Type Questions

25. Match the statements in **Column-I** with those in **Column-II**.

[IIT-JEE-2010 (Paper-2)]

[Note: Here z takes values in the complex plane and $\operatorname{Im} z$ and $\operatorname{Re} z$ denote, respectively, the imaginary part and the real part of z .]

Column I

Column II

- | | |
|--|---|
| (A) The set of points z satisfying $ z - i z = z + i z $ is contained in or equal to | (p) An ellipse with eccentricity $\frac{4}{5}$ |
| (B) The set of points z satisfying $ z + 4 + z - 4 = 10$ is contained in or equal to | (q) The set of points z satisfying $\operatorname{Im} z = 0$ |
| (C) If $ w = 2$, then the set of points $z = w - \frac{1}{w}$ is contained in or equal to | (r) The set of points z satisfying $ \operatorname{Im} z \leq 1$ |
| (D) If $ w = 1$, then the set of points $z = w + \frac{1}{w}$ is contained in or equal to | (s) The set of points z satisfying $ \operatorname{Re} z \leq 2$ |
| | (t) The set of points z satisfying $ z \leq 3$ |

26. Let $z_k = \cos\left(\frac{2k\pi}{10}\right) + i \sin\left(\frac{2k\pi}{10}\right); k = 1, 2, \dots, 9$.

[JEE (Adv)-2014 (Paper-2)]

List-I

- P. For each z_k there exists a z_j such that
 $z_k \cdot z_j = 1$
- Q. There exists a $k \in \{1, 2, \dots, 9\}$ such that
 $z_1 \cdot z = z_k$ has no solution z in the set of
 complex numbers
- R. $\frac{|1-z_1| |1-z_2| \dots |1-z_9|}{10}$ equals

List-II

1. True
2. False
3. 1
4. 2

S. $1 - \sum_{k=1}^9 \cos\left(\frac{2k\pi}{10}\right)$ equals

P	Q	R	S
(A) 1	2	4	3
(B) 2	1	3	4
(C) 1	2	3	4
(D) 2	1	4	3

Integer / Numerical Value Type Questions

27. Let ω be the complex number $\cos\frac{2\pi}{3} + i \sin\frac{2\pi}{3}$. Then the number of distinct complex numbers z satisfying

$$\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 0$$
 is equal to

[IIT-JEE-2010 (Paper-1)]

28. If z is any complex number satisfying $|z - 3 - 2i| \leq 2$, then the minimum value of $|2z - 6 + 5i|$ is

[IIT-JEE-2011 (Paper-1)]

29. Let $\omega = e^{i\pi/3}$, and a, b, c, x, y, z be non-zero complex number such that

$$\begin{aligned} a + b + c &= x \\ a + b\omega + c\omega^2 &= y \\ a + b\omega^2 + c\omega &= z \end{aligned}$$

Then the value of $\frac{|x|^2 + |y|^2 + |z|^2}{|a|^2 + |b|^2 + |c|^2}$ is

[IIT-JEE-2011 (Paper-2)]

30. For any integer k , let $\alpha_k = \cos\left(\frac{k\pi}{7}\right) + i \sin\left(\frac{k\pi}{7}\right)$, where $i = \sqrt{-1}$. The value of the expression

$$\frac{\sum_{k=1}^{12} |\alpha_{k+1} - \alpha_k|}{\sum_{k=1}^3 |\alpha_{4k-1} - \alpha_{4k-2}|}$$
 is

[JEE (Adv)-2015 (Paper-2)]

31. Let $\omega \neq 1$ be a cube root of unity. Then the minimum of the set $\{|a + b\omega + c\omega^2|^2 : a, b, c \text{ distinct non-zero integers}\}$ equals _____. **[JEE (Adv)-2019 (Paper-1)]**
32. For a complex number z , let $\operatorname{Re}(z)$ denote the real part of z . Let S be the set of all complex numbers z satisfying $z^4 - |z|^4 = 4iz^2$, where $i = \sqrt{-1}$. Then the minimum possible value of $|z_1 - z_2|^2$, where $z_1, z_2 \in S$ with $\operatorname{Re}(z_1) > 0$ and $\operatorname{Re}(z_2) < 0$, is _____. **[JEE (Adv)-2020 (Paper-2)]**
33. Let z be a complex number with non-zero imaginary part. If $\frac{2+3z+4z^2}{2-3z+4z^2}$ is a real number, then the value of $|z|^2$ is _____. **[JEE (Adv)-2022 (Paper-1)]**
34. Let $\bar{z}$ denote the complex conjugate of a complex number z and let $i = \sqrt{-1}$. In the set of complex numbers, the number of distinct roots of the equation $\bar{z} - z^2 = i(\bar{z} + z^2)$ is _____. **[JEE (Adv)-2022 (Paper-1)]**

□ □ □

